

Homework 8

4.581.)

a.

We want to prove that if P is an orthogonal projection matrix then so is $I - P$. To do this, we have to show

$$I - P = (I - P)^\top$$

$$(I - P)^2 = I - P$$

For the first condition,

$$(I - P)^\top = I^\top - P^\top = I - P$$

For the second condition,

$$(I - P)^2 = (I - P)(I - P) = I(I - P) - P(I - P) = I^2 - IP - PI + P^2 = I - P - P + P = I - P$$

Hence, if P is an orthogonal projection matrix then so is $I - P$.

b.

We want to prove that if the columns of $U \in \mathbb{R}^{m \times r}$ are orthonormal, then $UU^\top$ is an orthogonal projection matrix. To do this, we have to show

$$UU^\top = (UU^\top)^\top$$

$$(UU^\top)^2 = UU^\top$$

For the first condition,

$$(UU^\top)^\top = (U^\top)^\top (U)^\top = UU^\top$$

Note that if the columns of $U \in \mathbb{R}^{m \times r}$ are orthonormal then we have $U^\top U = I_r$. For the second condition,

$$(UU^\top)^2 = UU^\top UU^\top = UI_r U^\top = UU^\top$$

Hence, if the columns of $U \in \mathbb{R}^{m \times r}$ are orthonormal, then $UU^\top$ is an orthogonal projection matrix. Note that we have $\text{range}(UU^\top) = \text{range}(U)$ because

$$x = Uz \Rightarrow U^\top x = U^\top Uz = z \Rightarrow x = Uz = UU^\top x$$

We have that $UU^\top$ projects onto $\text{range}(UU^\top) = \text{range}(U)$, thus $UU^\top$ projects onto $\text{range}(U)$. We know that for arbitrary complementary projection matrix $Q = I - P$ for projection matrix P that $\text{range}(Q) = \text{null}(P)$. We can make a similar argument to the one before to show that $\text{null}(UU^\top) = \text{null}(U^\top)$. Therefore, the orthogonal projection matrix $I - UU^\top$ projects onto $\text{null}(UU^\top) = \text{null}(U^\top)$, thus $I - UU^\top$ projects onto $\text{null}(U^\top)$.

c.

We want to prove that if $A \in \mathbb{R}^{m \times n}$ is full rank with $n \leq m$, then $A(A^\top A)^{-1}A^\top$ is an orthogonal projection matrix. To do this, we have to show

$$A(A^\top A)^{-1}A^\top = (A(A^\top A)^{-1}A^\top)^\top$$

$$(A(A^\top A)^{-1}A^\top)^2 = A(A^\top A)^{-1}A^\top$$

Note that since the gram matrix of A is guaranteed to be symmetric i.e. $A^\top A = (A^\top A)^\top$, we have that the gram matrix inverse is guaranteed to be symmetric i.e. $(A^\top A)^{-1} = ((A^\top A)^{-1})^\top$. For the first condition,

$$(A(A^\top A)^{-1}A^\top)^\top = (A^\top)^\top ((A^\top A)^{-1})^\top (A)^\top = A(A^\top A)^{-1}A^\top$$

For the second condition,

$$(A(A^\top A)^{-1}A^\top)^2 = A(A^\top A)^{-1}A^\top A(A^\top A)^{-1}A^\top = A(A^\top A)^{-1}(A^\top A)(A^\top A)^{-1}A^\top = A(A^\top A)^{-1}A^\top$$

Hence, if $A \in \mathbb{R}^{m \times n}$ is full rank with $n \leq m$, then $A(A^\top A)^{-1}A^\top$ is an orthogonal projection matrix. Note that we have $\text{range}(A(A^\top A)^{-1}A^\top) = \text{range}(A)$ because

$$x = Az \Rightarrow A^\top x = A^\top Az \Rightarrow z = (A^\top A)^{-1}A^\top x \Rightarrow x = A(A^\top A)^{-1}A^\top x$$

So we have that $A(A^\top A)^{-1}A^\top$ projects onto $\text{range}(A(A^\top A)^{-1}A^\top) = \text{range}(A)$, thus $A(A^\top A)^{-1}A^\top$ projects onto $\text{range}(A)$.

d.

We want to prove that if $A \in \mathbb{R}^{m \times n}$ is full rank with $m \leq n$, then $A^\top(AA^\top)^{-1}A$ is an orthogonal projection matrix. To do this, we have to show

$$\begin{aligned} A^\top(AA^\top)^{-1}A &= (A^\top(AA^\top)^{-1}A)^\top \\ (A^\top(AA^\top)^{-1}A)^2 &= A^\top(AA^\top)^{-1}A \end{aligned}$$

Note that since the gram matrix of A is guaranteed to be symmetric i.e. $AA^\top = (AA^\top)^\top$, we have that the gram matrix inverse is guaranteed to be symmetric i.e. $(AA^\top)^{-1} = ((AA^\top)^{-1})^\top$. For the first condition,

$$(A^\top(AA^\top)^{-1}A)^\top = (A)^\top((AA^\top)^{-1})^\top(A^\top)^\top = A^\top(AA^\top)^{-1}A$$

For the second condition,

$$(A^\top(AA^\top)^{-1}A)^2 = A^\top(AA^\top)^{-1}AA^\top(AA^\top)^{-1}A = A^\top(AA^\top)^{-1}(AA^\top)(AA^\top)^{-1}A = A^\top(AA^\top)^{-1}A$$

Hence, if $A \in \mathbb{R}^{m \times n}$ is full rank with $m \leq n$, then $A^\top(AA^\top)^{-1}A$ is an orthogonal projection matrix. Note that we have $\text{range}(A^\top(AA^\top)^{-1}A) = \text{range}(A^\top)$ because

$$x = A^\top z \Rightarrow Ax = AA^\top z \Rightarrow z = (AA^\top)^{-1}Ax \Rightarrow x = A^\top(AA^\top)^{-1}Ax$$

So we have that $A^\top(AA^\top)^{-1}A$ projects onto $\text{range}(A^\top(AA^\top)^{-1}A) = \text{range}(A^\top)$, thus $A^\top(AA^\top)^{-1}A$ projects onto $\text{range}(A^\top)$.

e.

Let $A = U\Sigma V^\top$ be defined as in the problem. We can see that

$$A = U_1\Sigma_1V_1^\top$$

By observing which columns of U or V appear in the expression for A or $A^\top$ and using the orthonormality of the U and V blocks, we find that the four fundamental subspaces are

$$\text{range}(A) = \text{span}(U_1)$$

$$\text{range}(A^\top) = \text{span}(V_1)$$

$$\text{null}(A) = \text{span}(V_2)$$

$$\text{null}(A^\top) = \text{span}(U_2)$$

f.

We want to prove that $AA^\dagger$ and $A^\dagger A$ are orthogonal projection matrices. We find that

$$AA^\dagger = (U_1\Sigma_1V_1^\top)(V_1\Sigma_1^{-1}U_1^\top) = U_1(\Sigma_1\Sigma_1^{-1})U_1^\top = U_1U_1^\top$$

$$A^\dagger A = (V_1\Sigma_1^{-1}U_1^\top)(U_1\Sigma_1V_1^\top) = V_1(\Sigma_1^{-1}\Sigma_1)V_1^\top = V_1V_1^\top$$

Since U_1 and V_1 have orthonormal columns, our solution for part *b* proves that $AA^\dagger = U_1U_1^\top$ and $A^\dagger A = V_1V_1^\top$ must be orthogonal projection matrices. Further, using more results from part *b*, we know that $AA^\dagger = U_1U_1^\top$ will project onto $\text{range}(U_1) = \text{range}(A)$ and that $A^\dagger A = V_1V_1^\top$ will project onto $\text{range}(V_1) = \text{range}(A^\top)$.

g.

First, we want to prove that the definition holds for $\mathcal{X}_{ls}$. Start by decomposing y into its orthogonal projection onto $\text{range}(A)$ and its complement orthogonal projection

$$y = AA^\dagger y + (I - AA^\dagger)y$$

Where $AA^\dagger y \in \text{range}(A)$, as shown in part *f*, and $(I - AA^\dagger)y \in \text{null}(A^\top)$, which is a property of complementary orthogonal projections. Then for any $x \in \mathbb{R}^n$, we have

$$\|Ax - y\|^2 = \|Ax - (AA^\dagger y + (I - AA^\dagger)y)\|^2 = \|Ax - AA^\dagger y\|^2 + \|(I - AA^\dagger)y\|^2$$

because $Ax - AA^\dagger y \in \text{range}(A)$ is orthogonal to $(I - AA^\dagger)y \in \text{null}(A^\top)$. Therefore, choosing x to minimize $\|Ax - y\|^2$ is equivalent to choosing x to minimize $\|Ax - AA^\dagger y\|^2 + \|(I - AA^\dagger)y\|^2$. The second term doesn't depend on x , so when choosing x to minimize we just need x s.t.

$$\|Ax - AA^\dagger y\|^2 = 0 \Rightarrow Ax = AA^\dagger y$$

Hence, the definition holds for $\mathcal{X}_{ls}$ i.e.

$$\mathcal{X}_{ls} = \{x | Ax = AA^\dagger y\} = \arg \min_x \|Ax - y\|$$

Further, for any $x \in X_{ls}$ we have

$$\|Ax - y\|^2 = \|Ax - AA^\dagger y\|^2 + \|(I - AA^\dagger)y\|^2 = 0 + \|(I - AA^\dagger)y\|^2 = \|(I - AA^\dagger)y\|^2$$

Thus, all least-square minimizers have the same residual $(I - AA^\dagger)y$.

h.

Consider the solution $x' = A^\dagger y$. Then $Ax' = AA^\dagger y$, so $x' \in X_{ls}$. Every other solution $x \in X_{ls}$ must satisfy $A(x - x') = 0$, or in other words, $x - x' \in \text{null}(A)$. In part *e*, we determined that

$$\text{null}(A) = \text{span}(V_2)$$

and using part *b*, we can see that $V_2 V_2^\top$ projects onto $\text{range}(V_2) = \text{null}(A)$. Notice that

$$V_2 V_2^\top = I - V_1 V_1^\top = I - A^\dagger A$$

We then have $\text{range}(I - A^\dagger A) = \text{range}(V_2 V_2^\top) = \text{range}(V_2) = \text{null}(A)$. Since $\text{null}(A) = \text{range}(I - A^\dagger A)$, we can rewrite $x - x' = (I - A^\dagger A)z$ for some $z \in \mathbb{R}^n$. Hence, every least-squares minimizer has the form

$$x = A^\dagger y + (I - A^\dagger A)z$$

for some $z \in \mathbb{R}^n$.

i.

Let x be an arbitrary element of X_{ls} s.t. $x \neq x_{\text{pinv}}$. Using part *h*, we can rewrite x as

$$x = A^\dagger y + (I - A^\dagger A)z = x_{\text{pinv}} + (I - A^\dagger A)z$$

for some $z \in \mathbb{R}^n$. Since $x \neq x_{\text{pinv}}$, we have $(I - A^\dagger A)z \neq 0$. Note that

$$x_{\text{pinv}} = A^\dagger y = V_1(\Sigma_1^{-1}U_1^\top y) \in \text{range}(A^\top) = \text{span}(V_1),$$

as follows directly from the SVD. Also, $(I - A^\dagger A)z \in \text{null}(A) = \text{span}(V_2)$, as shown in part *h*, and these two subspaces are orthogonal. Therefore

$$\begin{aligned} (I - A^\dagger A)z \neq 0 &\Rightarrow \|(I - A^\dagger A)z\|^2 > 0 \\ \|x\|^2 &= \|A^\dagger y\|^2 + \|(I - A^\dagger A)z\|^2 > \|A^\dagger y\|^2 = \|x_{\text{pinv}}\|^2 \end{aligned}$$

Hence, $x_{\text{pinv}} = A^\dagger y$ is the unique minimum-norm least-squares solution.

16.2560.)

a.

Our goal is to choose w to minimize $\|w\|$ subject to the constraint

$$\frac{1}{T} \sum_{t=1}^T (w^\top y_t)^2 = 1$$

We can rewrite the constraint as a quadratic form

$$\frac{1}{T} \sum_{t=1}^T (w^\top y_t)^2 = \frac{1}{T} \sum_{t=1}^T \|y_t^\top w\|^2 = \frac{1}{T} \sum_{t=1}^T (y_t^\top w)^\top y_t^\top w = \frac{1}{T} \sum_{t=1}^T w^\top y_t y_t^\top w = w^\top R w$$

where we define

$$R = \frac{1}{T} \sum_{t=1}^T y_t y_t^\top \in \mathbb{R}^{n \times n}$$

Note that minimizing $\|w\|$ is equivalent to minimizing $\|w\|^2 = w^\top w$. Therefore, an equivalent goal is to choose w to minimize $w^\top w$ subject to the constraint $w^\top R w = 1$. To do so, we can use a Lagrangian

$$\begin{aligned} L(w, \lambda) &= w^\top w - \lambda(w^\top R w - 1) \\ \nabla_\lambda L &= -(w^\top R w - 1) = 0 \Rightarrow w^\top R w = 1 \\ \nabla_w L &= 2w - 2\lambda R w = 0 \Rightarrow \frac{1}{\lambda} w = R w \end{aligned}$$

Solving the Lagrangian tells us that w must be an eigenvector of R . This reduces the problem to choosing the optimal eigenvector of R . Consider some arbitrary eigenvector v and corresponding eigenvalue λ_v of R . Then

$$v^\top R v = v^\top \lambda_v v = \lambda_v v^\top v = 1 \Rightarrow v^\top v = \frac{1}{\lambda_v}$$

Let u be the unit eigenvector of R corresponding to the largest positive eigenvalue. To minimize $\|w\|$ subject to the constraint above, we choose

$$w = \frac{u}{\sqrt{\lambda_u}}$$

so that

$$\begin{aligned} w^\top R w &= \frac{u^\top}{\sqrt{\lambda_u}} R \frac{u}{\sqrt{\lambda_u}} = \frac{1}{\lambda_u} u^\top \lambda_u u = u^\top u = 1 \\ w^\top w &= \frac{u^\top}{\sqrt{\lambda_u}} \frac{u}{\sqrt{\lambda_u}} = \frac{1}{\lambda_u} \leq \frac{1}{\lambda_i} \end{aligned}$$

for all other eigenvalues, λ_i , of R . Note that we can do this because R is symmetric and positive semidefinite. R is symmetric because it is a sum of symmetric matrices:

$$\begin{aligned} (y_t y_t^\top)^\top &= y_t y_t^\top \\ R &= \frac{1}{T} \sum_{t=1}^T y_t y_t^\top \end{aligned}$$

R is positive semidefinite by construction. For any $x \in \mathbb{R}^n$,

$$x^\top R x = \frac{1}{T} \sum_{t=1}^T x^\top (y_t y_t^\top) x = \frac{1}{T} \sum_{t=1}^T (x^\top y_t)^2 \geq 0.$$

Since R is symmetric, we can diagonalize it to find its eigenvectors and eigenvalues (which are real), and since R is positive semidefinite, it has positive eigenvalues.

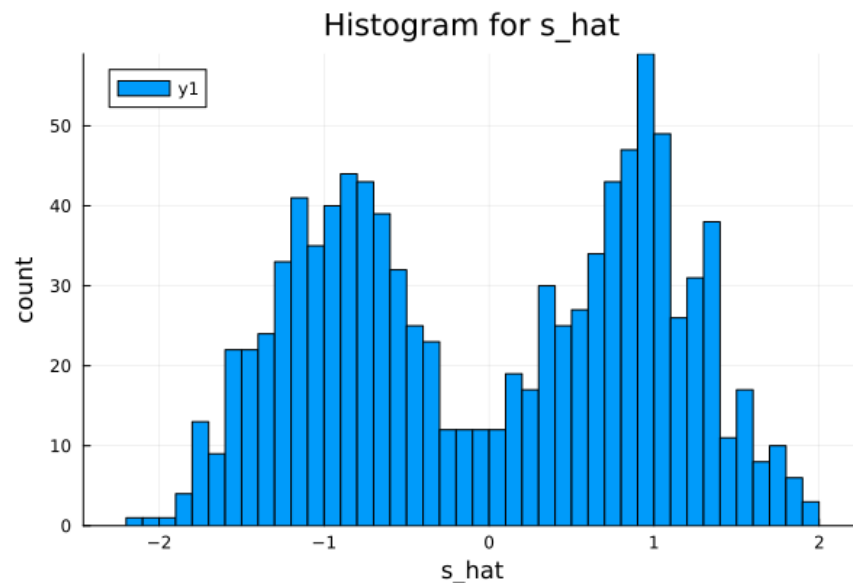
b.

This question was done in Julia, the code and plot can be seen below. The weight vector w and the error rate are in the output of the code below, it can be seen that the error rate is 0.029.

```

1  using LinearAlgebra, Plots, Statistics
2
3  #Question 2.)
4
5  include("readclassjson.jl")
6  data = readclassjson("bs_det_data.json")
7
8
9  Y = data["Y"]
10 s = data["s"]
11 T = data["T"]
12 n = data["n"]
13 R = (1/T) * (Y * Y')
14
15 eig = eigen(R)
16 eigenvalues = eig.values
17 eigenvectors = eig.vectors
18 i_max = argmax(eigenvalues)
19 u = eigenvectors[:, i_max]
20 lambda_u = eigenvalues[i_max]
21
22 w = -u / sqrt(lambda_u)
23 println("Weight vector w:")
24 println(w)
25
26 s_hat = w' * Y
27 histogram(vec(s_hat), bins=60,
28          xlabel="s_hat",
29          ylabel="count",
30          title="Histogram for s_hat")
31 savefig("hw8_q2b.png")
32
33 s_est = map(x -> x >= 0 ? 1 : -1, vec(s_hat))
34 err = mean(s_est .!= s)
35 println("\nError rate = ", err)

```



Weight vector w:
 [-0.03224530108439294, -0.09600678070245434, 0.012300376696346484, 0.019867563858643623, -0.06334718004403021, 0.0662685979325205, 0.07769274840286249, -0.003748387075843224, 0.01651599450852731, 0.012568547850148517, -0.013991403119214395, 0.03722671802828625, -0.03154105297560477, 0.12525945464545402, -0.013700224050555125, 0.01046626044477991, 0.060617342792573393, 0.010108430511189832, -0.009208138818350375, -0.04747941079048275]

Error rate = 0.029

16.2670.)

a.

We've previously solved for the minimizer of a weighted sum objective

$$x_{\text{reg}} = \arg \min_x \|Ax - y\|^2 + \mu \|x\|^2$$

$$x_{\text{reg}} = (A^\top A + \mu I)^{-1} A^\top y$$

Thus if $x_{\text{reg}} = By$, then

$$B = (A^\top A + \mu I)^{-1} A^\top$$

Let $A = U\Sigma V^\top$ be the SVD of A where $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$, $U = [u_1 \ \dots \ u_n]$, $V = [v_1 \ \dots \ v_n]$. Then we have

$$A^\top = (U\Sigma V^\top)^\top = V\Sigma^\top U^\top = V\Sigma U^\top$$

$$A^\top A = V\Sigma U^\top U\Sigma V^\top = V\Sigma^2 V^\top$$

Therefore,

$$\begin{aligned} B &= (V\Sigma^2 V^\top + \mu I)^{-1} V\Sigma U^\top = (V(\Sigma^2 + \mu I)V^\top)^{-1} V\Sigma U^\top \\ &= (V^\top)^{-1} (\Sigma^2 + \mu I)^{-1} V^{-1} V\Sigma U^\top = V(\Sigma^2 + \mu I)^{-1} \Sigma U^\top \end{aligned}$$

Note that we have

$$\begin{aligned} \Sigma^2 + \mu I &= \text{diag}(\sigma_1^2 + \mu, \dots, \sigma_n^2 + \mu) \\ (\Sigma^2 + \mu I)^{-1} &= \text{diag}\left(\frac{1}{\sigma_1^2 + \mu}, \dots, \frac{1}{\sigma_n^2 + \mu}\right) \\ (\Sigma^2 + \mu I)^{-1} \Sigma &= \text{diag}\left(\frac{\sigma_1}{\sigma_1^2 + \mu}, \dots, \frac{\sigma_n}{\sigma_n^2 + \mu}\right) \end{aligned}$$

We can then rewrite B as a summation of matrices

$$B = \sum_{i=1}^n \frac{\sigma_i}{\sigma_i^2 + \mu} v_i u_i^\top$$

We can now define $\tilde{\sigma}_i$, $\tilde{u}_i$ and $\tilde{v}_i$ for $i = 1, \dots, n$

$$\begin{aligned} \tilde{\sigma}_i &= \frac{\sigma_i}{\sigma_i^2 + \mu} \\ \tilde{u}_i &= v_i \\ \tilde{v}_i &= u_i \end{aligned}$$

Then we have

$$B = \sum_{i=1}^n \tilde{\sigma}_i \tilde{u}_i \tilde{v}_i^\top = \sum_{i=1}^n \frac{\sigma_i}{\sigma_i^2 + \mu} v_i u_i^\top$$

Note that $\sigma_i \geq \sigma_j \not\Rightarrow \frac{\sigma_i}{\sigma_i^2 + \mu} \geq \frac{\sigma_j}{\sigma_j^2 + \mu}$. To maintain the convention $\tilde{\sigma}_1 \geq \dots \geq \tilde{\sigma}_n$, we may need to permute the indices so that the singular values are sorted in decreasing order, and reorder the corresponding singular vectors $\tilde{u}_i$ and $\tilde{v}_i$ accordingly.

b.

The operator norm of B is the largest singular value, $\tilde{\sigma}_1$, of B . From part *a*, we have

$$\|B\| = \tilde{\sigma}_1 = \max_{i \in \{1, \dots, n\}} \frac{\sigma_i}{\sigma_i^2 + \mu}$$

where σ_i are the singular values of A .

c.

We can rewrite the formula for the worst-case relative inversion error

$$\max_{y \neq 0} \frac{\|AB y - y\|}{\|y\|} = \max_{y \neq 0} \frac{\|(AB - I)y\|}{\|y\|} = \|AB - I\|$$

In other words, the worst-case relative inversion error is the operator norm of $AB - I$. Using the SVDs of A and B as defined in part *a*, we have

$$AB = U \Sigma V^\top V (\Sigma^2 + \mu I)^{-1} \Sigma U^\top = U \Sigma (\Sigma^2 + \mu I)^{-1} \Sigma U^\top = \sum_{i=1}^n \frac{\sigma_i^2}{\sigma_i^2 + \mu} u_i u_i^\top$$

Then

$$AB - I = U \Sigma (\Sigma^2 + \mu I)^{-1} \Sigma U^\top - U I U^\top = U (\Sigma (\Sigma^2 + \mu I)^{-1} \Sigma - I) U^\top = \sum_{i=1}^n \left(\frac{\sigma_i^2}{\sigma_i^2 + \mu} - 1 \right) u_i u_i^\top = \sum_{i=1}^n \frac{-\mu}{\sigma_i^2 + \mu} u_i u_i^\top$$

Similar to the end of part *a*, we order by magnitude to preserve convention, and define $\frac{\mu}{\sigma_i^2 + \mu}$ to be the singular values of $AB - I$. Note that singular values can't be negative, so we move those negatives to the vectors instead, which is why we have positive μ in the numerator. We can then use the singular values of $AB - I$ to find the worst-case relative inversion error

$$\max_{y \neq 0} \frac{\|AB y - y\|}{\|y\|} = \|AB - I\| = \max_{i \in \{1, \dots, n\}} \frac{\mu}{\sigma_i^2 + \mu} = \frac{\mu}{\sigma_n^2 + \mu}$$

This holds because

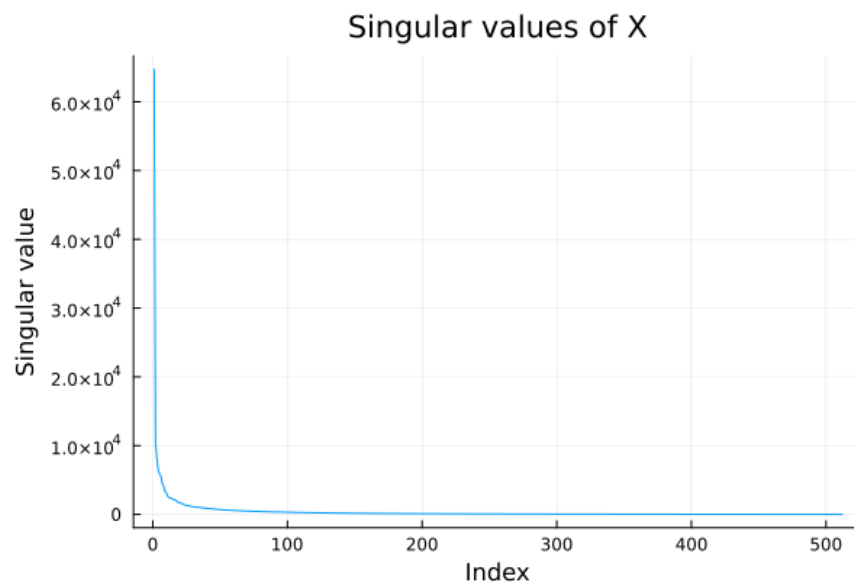
$$\sigma_1 \geq \dots \geq \sigma_n \Rightarrow \frac{\mu}{\sigma_n^2 + \mu} \geq \dots \geq \frac{\mu}{\sigma_1^2 + \mu}$$

16.2900.)

a.

This question was done in Julia, the code and plot can be seen below. As can be seen in the plot below, the singular values of X decrease very rapidly. A small percentage of the singular values provide significant impact to the matrix, while the remaining singular values are comparatively very small and unimportant. This suggests that the image is highly compressible using a low-rank approximation.

```
1  using LinearAlgebra, Plots, Statistics
2
3  #Question 4.)
4
5  include("readclassjson.jl")
6  data = readclassjson("lena_data.json")
7  X = data["X"]
8
9  #Part a
10 U, S, V = svd(X)
11
12 # Plot singular values
13 plot(S, xlabel="Index",
14      ylabel="Singular value",
15      title="Singular values of X",
16      label="")
17 savefig("hw8_q4a.png")
```



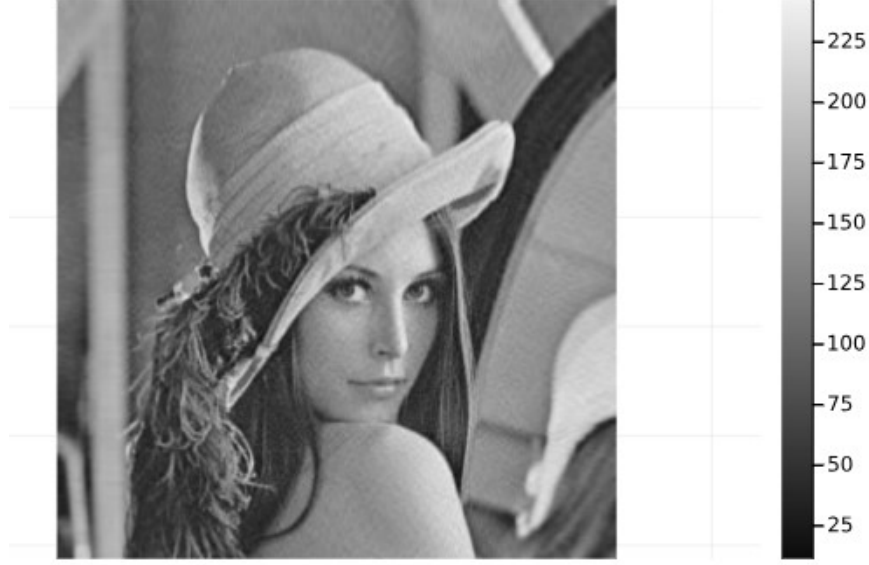
b.

This question was done in Julia, the code, output and explanation can be seen below.

```
19 #Part b
20 k = 75
21 X_tilde = U[:, 1:k] * Diagonal(S[1:k]) * V[:, 1:k]'
22
23 # Frobenius norm relative error
24 rel_error = norm(X - X_tilde) / norm(X)
25 println("Relative Frobenius norm error: ", rel_error)
26
27 # Display low-rank approximation
28 heatmap(X_tilde, c=:grays, yflip=true, axis=false, aspect_ratio=:equal)
```



```
Relative Frobenius norm error: 0.042386936699369944
```



As can be seen in the output of the code above,

$$\frac{\|X - \tilde{X}\|_F}{\|X\|_F} = 0.0424$$

The low rank approximation, $\tilde{X}$, can be represented as

$$\tilde{X} = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$$

where $\tilde{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_{75})$, $\tilde{U} = [u_1 \cdots u_{75}]$, $\tilde{V} = [v_1 \cdots v_{75}]$. In other words, $\tilde{U}$ and $\tilde{V}$ preserve the first 75 columns of U and V respectively and $\tilde{\Sigma}$ preserves the largest 75 singular values in Σ . This means we have $\tilde{\Sigma} \in \mathbb{R}^{75 \times 75}$ and $\tilde{U}, \tilde{V} \in \mathbb{R}^{512 \times 75}$. To find how many doubles it takes to store the low rank approximation, we want to find how many doubles it takes to store $\tilde{\Sigma}$, $\tilde{U}$ and $\tilde{V}$. As $\tilde{U}, \tilde{V} \in \mathbb{R}^{512 \times 75}$, they both require 512×75 doubles to store. Since $\tilde{\Sigma} \in \mathbb{R}^{75 \times 75}$ is a diagonal matrix, we only need to store its diagonal entries, so it requires 75 doubles to store. Thus, the total number of doubles required to store the low rank approximation $\tilde{X}$ is

$$2(512 \times 75) + 75 = 76875$$

Therefore, the compression ratio of the low rank approximation with respect to the original 512^2 doubles is

$$\frac{512^2}{76875} = 3.41$$

This means the low rank approximation uses about $3.4 \times$ less storage than the original image.

17.940.)

a.

Let x, y and z be defined as they are in the problem. As x is uniform, the pdf of x is

$$f_x(x) = \begin{cases} \frac{1}{1-(-1)} = \frac{1}{2}, & x \in [-1, 1] \\ 0, & \text{otherwise} \end{cases}$$

Similarly, the pdf of y is

$$f_y(y) = \begin{cases} \frac{1}{1-(-1)} = \frac{1}{2}, & y \in [-1, 1] \\ 0, & \text{otherwise} \end{cases}$$

The joint pdf of independent variables is the product of the marginals. So if $x, y \in [-1, 1]$ then

$$f_z(x, y) = f_x(x)f_y(y) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Otherwise either $x \notin [-1, 1]$ or $y \notin [-1, 1]$, so $f_z(x, y) = 0$. Hence, the pdf of z is

$$f_z(x, y) = \begin{cases} \frac{1}{4}, & x, y \in [-1, 1] \\ 0, & \text{otherwise} \end{cases}$$

b.

Let p, q and w be defined as they are in the problem. As p is Gaussian, the pdf of p is

$$f_p(p) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{p-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}p^2}$$

Similarly, the pdf of q is

$$f_q(q) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{q-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}q^2}$$

The joint pdf of independent variables is the product of the marginals. Hence, the pdf of w is

$$f_w(p, q) = f_p(p)f_q(q) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}p^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}q^2} = \frac{1}{2\pi} e^{-\frac{1}{2}(p^2+q^2)}$$

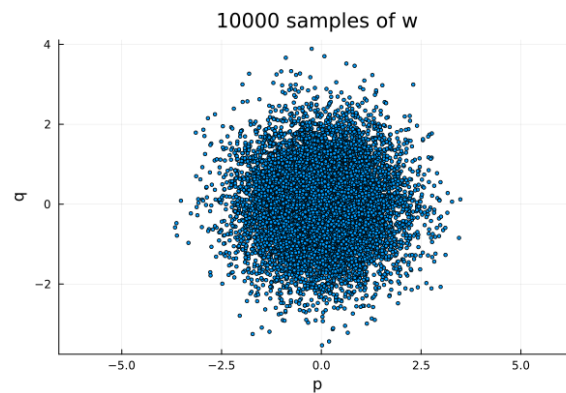
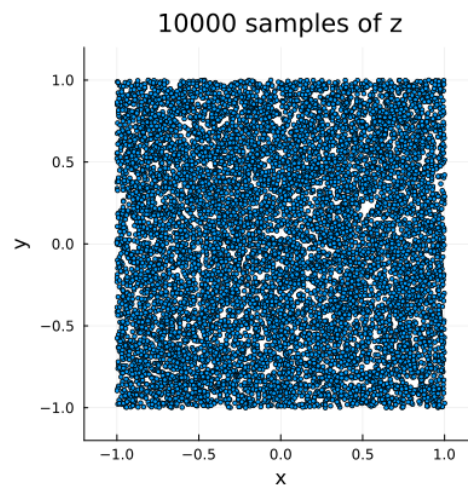
c.

This question was done in Julia, the code and plots can be seen below.

```

1  using Distributions, Plots
2
3  # Question 5
4
5  N = 10_000
6
7  x_samples = rand(Uniform(-1, 1), N)
8  y_samples = rand(Uniform(-1, 1), N)
9  plot_z = scatter(
10     x_samples, y_samples,
11     markersize = 2,
12     aspect_ratio = :equal,
13     title = "10000 samples of z",
14     xlabel = "x",
15     ylabel = "y",
16     legend = false,
17     xlim = (-1.2, 1.2),
18     ylim = (-1.2, 1.2)
19 )
20 savefig("hw8_q5ci.png")
21
22 p_samples = rand(Normal(0, 1), N)
23 q_samples = rand(Normal(0, 1), N)
24 plot_w = scatter(
25     p_samples, q_samples,
26     markersize = 2,
27     aspect_ratio = :equal,
28     title = "10000 samples of w",
29     xlabel = "p",
30     ylabel = "q",
31     legend = false
32 )
33 savefig("hw8_q5cii.png")

```



d.

Let r and T be as the problem described and w be as part b described. Then,

$$T = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \Rightarrow \det(T) = \cos^2 \theta + \sin^2 \theta = 1$$

Since $|\det(T)| = 1$, this means that T is invertible, so $w = T^{-1}r$. Further,

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = I \Rightarrow T^{-1} = T^\top = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

So T^{-1} is a rotation matrix as well. Because r is a linear change of variables of w , we have

$$f_r(r) = \frac{1}{|\det(T)|} f_w(T^{-1}r) = f_w(T^{-1}r)$$

Note that for $w = \begin{bmatrix} p \\ q \end{bmatrix}$, we have $\|w\|^2 = p^2 + q^2$, so we can rewrite the pdf of w to be

$$f_w(w) = \frac{1}{2\pi} e^{-\frac{1}{2}\|w\|^2}$$

We've proved previously that orthogonal matrices preserve vector norm i.e. for any orthogonal matrix U and vector x , $\|Ux\| = \|x\|$. Since rotation matrices are orthogonal, and T^{-1} is a rotation matrix, we have $\|T^{-1}r\| = \|r\|$. Therefore,

$$f_r(r) = f_w(T^{-1}r) = \frac{1}{2\pi} e^{-\frac{1}{2}\|T^{-1}r\|^2} = \frac{1}{2\pi} e^{-\frac{1}{2}\|r\|^2} = f_w(r)$$

We thus have $f_r(r) = f_w(r)$, so r and w have the same pdf. Hence, w is circularly symmetric.